Critical Mineral Company Stocks: VMD-Based Forecasting and Systematic Trading

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Abstract

This report walks through a highly practical, data-driven approach for predicting and trading Rare Earth Minerals. Because these commodity markets can swing wildly and behave unpredictably, we use a "divide-and-conquer" method. First, we break the raw daily price data into simpler, smoother trends and faster, noisier movements using a technique called Variational Mode Decomposition (VMD). We then use a machine learning tool (LASSO) to throw out any irrelevant data, making sure we only rely on the most useful indicators for forecasting. To actually predict the prices, we match the simple, clean trends to a basic statistical model, but handle the messy, chaotic price noise using an advanced deep-learning neural network (LSTM). Finally, we use an Interval Multi-Layer Perceptron to predict sensible "High" and "Low" boundaries around our forecast. By only trading when our predicted return sits comfortably inside these confidence bounds, we can build a much safer trading strategy with fewer devastating drawdowns and better overall returns.

1 1. Introduction

Rare earth minerals are critical inputs for modern electronics and renewable energy infrastructure. Due to concentrated supply chains and geopolitical factors, their market prices are highly volatile. Accurate price forecasting is therefore essential for both industrial procurement and financial trading.

Predicting these prices is challenging because traditional statistical models struggle with sudden market shifts, while deep neural networks often overfit to noise. To address this, we adapted a multimodal forecasting framework. Rather than predicting the raw price directly, we decompose the historical price data into simpler frequency components. We then incorporate external variables—such as energy prices, stock indices, and Google Trends data—to improve predictive accuracy. Finally, to evaluate the practical value of these forecasts, we test them in a daily trading simulation.

2 2. Data and Methodology

In order to accurately forecast rare earth index prices and formulate a practical trading strategy, we construct a hybrid framework consisting of data decomposition, feature selection, point forecasting, and interval forecasting. We benchmark our deep learning techniques against traditional statistical and machine learning approaches.

2.1 2.2 VMD Decomposition and IMF Feature Engineering

VMD is an adaptive signal processing algorithm that decomposes a non-stationary signal $f(t)$ into K discrete amplitude-modulated and frequency-modulated sub-signals (modes), $u_k(t)$. The goal is to minimize the sum of the estimated bandwidths of the modes. The constrained variational

problem is given by:

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \tag{1}$$

subject to $\sum_{k=1}^{K} u_k(t) = f(t)$, where ω_k is the center frequency of the k-th mode, and $\delta(t)$ is the Dirac distribution.

2.2 2.3 Approximate Entropy (AE)

To quantify the complexity of each decomposed mode, we rely on Approximate Entropy (ApEn). A lower ApEn indicates a predictable, deterministic trend, while higher ApEn scores imply noisy, chaotic behavior. For a sequence length N, given embedding dimension m and tolerance r, it is defined as:

$$ApEn(m, r, N) = \Phi^m(r) - \Phi^{m+1}(r) \tag{2}$$

where $\Phi^m(r)$ is the average of the logarithmic probabilities that two sequences patterns that are close within m points remain close for $m + 1$ points.

2.3 2.4 Least Absolute Shrinkage and Selection Operator (LASSO) Regression

To eliminate noisy, unhelpful predictors and prevent overfitting, we apply the Least Absolute Shrinkage and Selection Operator (LASSO) to the external variables for each respective mode. LASSO applies an L_1 penalty to the regression coefficients:

$$\min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^{N} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\} \tag{3}$$

where λ controls the shrinkage intensity. Coefficients of irrelevant features are forced exactly to zero.

2.4 2.7 Benchmark Models and Parameter Settings

Based on the ApEn scores, the categorized modes undergo unique prediction architectures:

- **Low-Complexity Modes (ARX):** Smooth trends are forecast directly using an Autoregressive Exogenous (ARX) model, capturing linear relationships with predictors.

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \beta_j x_{t-j} + \epsilon_t \tag{4}$$

- **High-Complexity Modes (LSTM):** Noisy variations are processed through Long Short-Term Memory networks. LSTMs solve the vanishing gradient problem using gated cell states. The fundamental memory updates are governed by:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget gate)} \tag{5}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input gate)} \tag{6}$$
$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{7}$$
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad \text{(Cell state update)} \tag{8}$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{9}$$
$$h_t = o_t * \tanh(C_t) \tag{10}$$

Benchmark Models: To strictly evaluate performance, our combined forecast is measured against:

1. **ARIMA**: A purely statistical time-series baseline estimating future returns through autoregressive and moving average properties $ARIMA(p, d, q)$.
2. **SVR (Support Vector Regression)**: Maps data points into high-dimensional space via an ϵ-insensitive loss function loss to fit non-linear curves.
3. **BPNN (Backpropagation Neural Network)**: A traditional Multi-Layer Perceptron utilizing iterative weight gradient adjustments, acting as a standard neural baseline.

2.5 2.9 Trading Strategy Design

Unlike point forecasts that simply predict a singular next-day price P_{t+1}, we apply an Interval Neural Network (MLP) trained specifically to generate Lower (L_t) and Upper (U_t) confidence bounds using a specialized coverage width-based loss function. A directional trade (buy or short) factoring in a transaction cost c is simulated. In the standard system, a position is entered simply if predicted directional return beats c. In our specialized **Interval Trading** framework, trade execution is prohibited if the predicted point target falls outside the produced safety bounds:

$$\text{Trade}_{t+1} = \begin{cases} +1 \text{ (Long)} & \text{if } \hat{P}_{t+1} > P_t(1+c) \text{ and } \hat{P}_{t+1} \in [L_t, U_t] \\ -1 \text{ (Short)} & \text{if } \hat{P}_{t+1} < P_t(1-c) \text{ and } \hat{P}_{t+1} \in [L_t, U_t] \\ 0 \text{ (Hold)} & \text{Otherwise} \end{cases} \tag{11}$$

2.6 2.1 Sample and Data Sources

To capture exactly what drives rare earth prices, we combined standard financial history with alternative behavioral data, making sure everything lined up perfectly date-by-date.

2.6.1 The Target: Rare Earth Index

Our main target is the **Rare Earth Index**. We built this by taking the daily closing stock prices of 8 major global rare earth mining and production companies and averaging them into a single trackable index. This serves as a reliable pulse of the overall industry.

2.6.2 Standard Financial Indicators

Global economic conditions heavily influence material costs. To map these forces, we rely on established macroeconomic dependencies [**?**, **?**, **?**, **?**, **?**]:

- **Shanghai Composite Index**: China controls approximately 70% of global rare earth extraction and over 90% of refining capabilities. Therefore, Chinese domestic economic health and manufacturing quotas (proxied by the Shanghai Index) are the primary anchors for global rare earth pricing algorithms, vastly superseding extraction-only indices like the ASX 200 [**?**].
- **Crude Oil WTI**: Extracting and separating these metals is highly energy-intensive. Furthermore, crude oil price spikes accelerate the global transition to renewable energy (EVs and wind turbines), which are the primary consumers of rare earth permanent magnets [**?**].
- **S&P 500 & VIX**: Rare earths are strategic assets sensitive to global trade tensions. The VIX (volatility index) captures supply chain fears, while the S&P 500 proxies the purchasing power of Western technology manufacturers [**?**].
- **US Dollar Index**: Because rare earths are internationally traded commodities modeled in USD, currency fluctuations inversely impact their global purchasing power [**?**].

2.6.3 Public Sentiment

Financial markets aren't purely logical; they are heavily influenced by hype, news cycles, and investor attention.

- **Google Trends Search Volume**: We pulled the total worldwide search volume for the keyword "*rare earth*".
- **News-Based Economic Sentiment Index (NESI)**: Developed by the Federal Reserve Bank of San Francisco, this index uses natural language processing to scan hundreds of thousands of financial news articles daily to assess real-time market optimism and pessimism.

Table 1: Data Description and Sourcing

Variable	Description
Target_Close	Individual Rare Earth Company Stock Close
SP500	S&P 500 Index
Shanghai_Index	Shanghai Composite
Crude_Oil	WTI Crude Oil Futures
USD_CNY	USD/CNY Exchange Rate
Search_Index	Google Trends (worldwide)
News_Sentiment	NESI Score (Fed SF)

Table 2: Descriptive Statistics of Market Data

EnergyFuels

Mean	Max	Min	Std	Obs
18.0368	234.8800	0.5000	34.4284	4880

NeoPerformance

Mean	Max	Min	Std	Obs
12.0247	20.6377	4.0527	3.0352	4880

Shenghe

Mean	Max	Min	Std	Obs
1.4060	4.8950	0.1620	0.8161	4880

Lynas

Mean	Max	Min	Std	Obs
4.7731	26.9180	0.2210	4.5701	4880

Arafura

Mean	Max	Min	Std	Obs
0.2534	1.6620	0.0280	0.3098	4880

Iluka

Mean	Max	Min	Std	Obs
4.0522	10.8600	0.8920	1.9495	4880

MPMaterials

Mean	Max	Min	Std	Obs
16.1475	98.6500	9.7800	12.7781	4880

ChinaNorthern

Mean	Max	Min	Std	Obs
2.5205	9.4680	0.0560	1.7071	4880

Figure 1: Raw Rare Earth Index Price Over Time

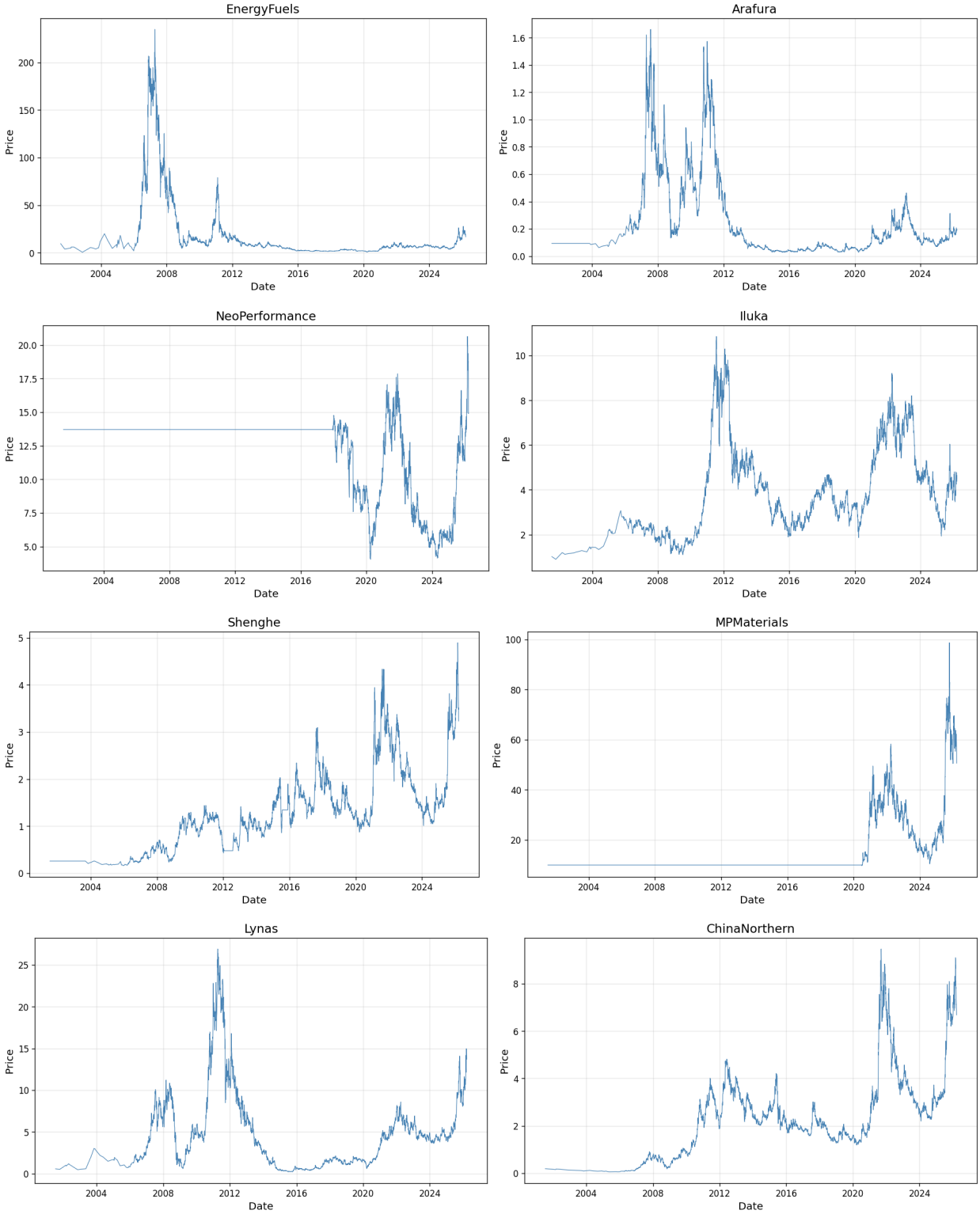
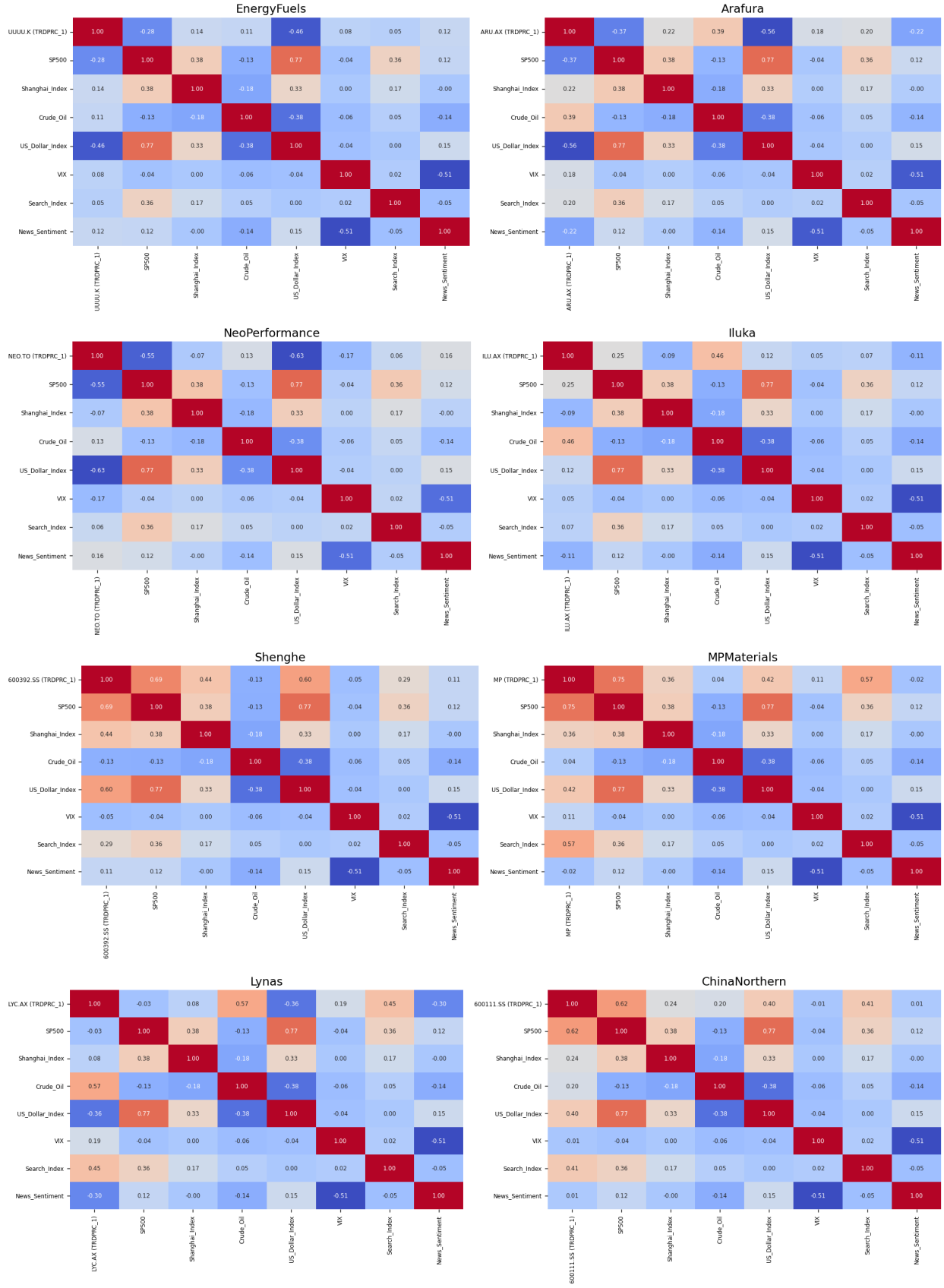


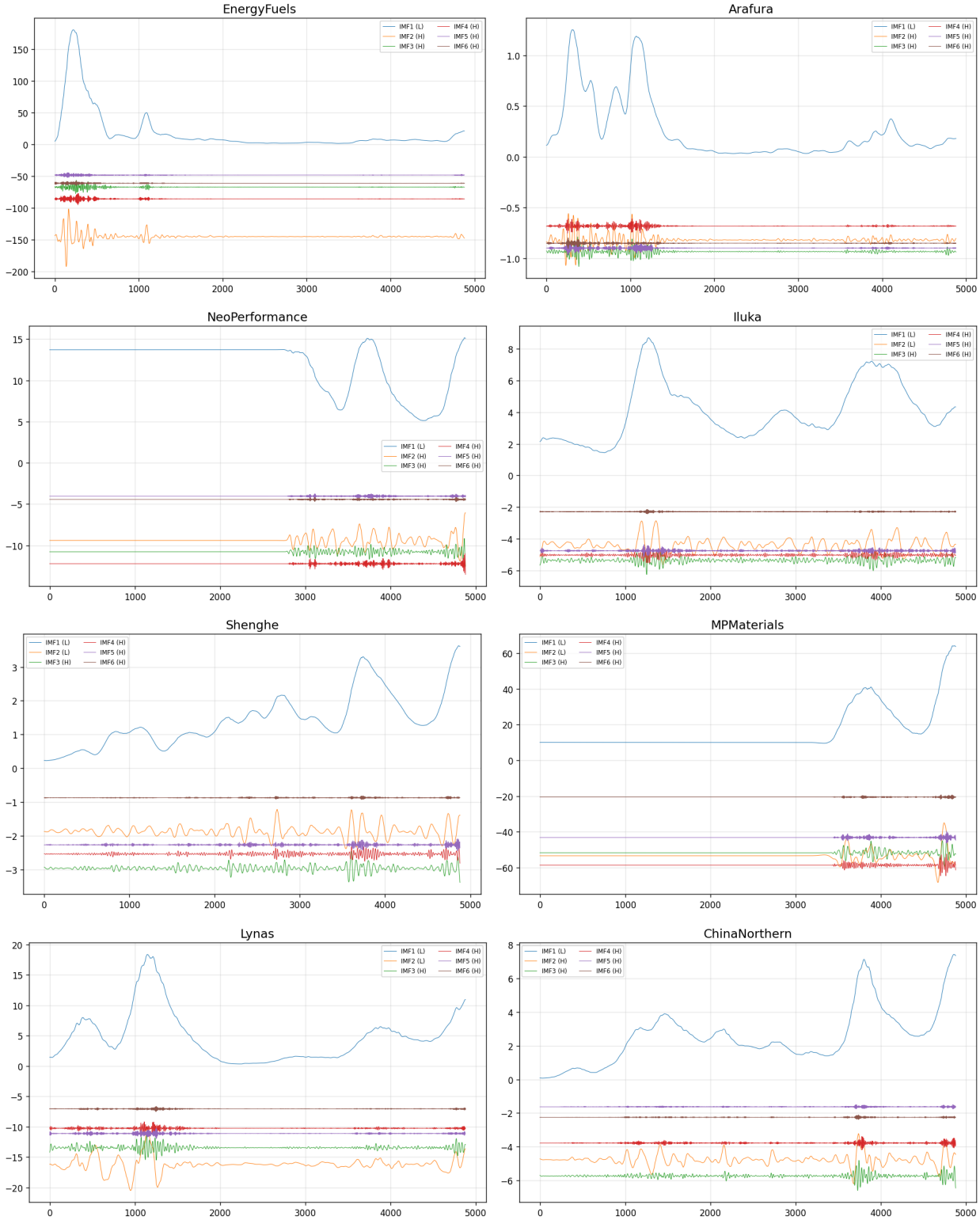
Figure 2: A Closer Look at the Volatility (2021)

Figure 3: Correlation Heatmap Showing How Closely the Inputs Move Together



Trying to predict a jagged, volatile chart is hard. So, we use Variational Mode Decomposition (VMD) to cleanly slice the main Rare Earth Index into four separate waves (or modes). Think of it like taking a messy musical chord and isolating the simple individual notes.

Figure 4: Separating the Target Price into Four Manageable Modes



Once we have our four separate waves, we need to decide how to treat them. We measure the "chaos" of each wave using a metric called Approximate Entropy (ApEn). If a wave has a low ApEn score, it means it's a smooth, highly predictable long-term trend. If it has a high ApEn score, it represents fast, unpredictable noise. We categorize the waves based on this score so we can match them to the right forecasting tool later.

Table 3: Categorized Separated Modes (VMD Entropy Grid)

EnergyFuels

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.012	85.6	0.946	0.984	0.011	Low
IMF2	0.019	51.4	0.023	0.215	0.121	High
IMF3	0.062	16.0	0.001	0.066	0.238	High
IMF4	0.114	8.8	0.001	0.046	0.258	High
IMF5	0.277	3.6	0.000	0.023	0.157	High
IMF6	0.390	2.6	0.000	0.018	0.255	High

Arafura

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.008	122.0	0.935	0.981	0.015	Low
IMF2	0.018	56.1	0.022	0.222	0.211	High
IMF3	0.036	27.4	0.004	0.117	0.304	High
IMF4	0.079	12.6	0.001	0.065	0.380	High
IMF5	0.127	7.8	0.001	0.045	0.361	High
IMF6	0.240	4.2	0.000	0.030	0.297	High

NeoPerformance

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.206	4.8	0.913	0.976	0.010	Low
IMF2	0.212	4.7	0.032	0.285	0.155	High
IMF3	0.276	3.6	0.005	0.115	0.293	High
IMF4	0.314	3.2	0.001	0.056	0.289	High
IMF5	0.393	2.5	0.000	0.026	0.288	High
IMF6	0.469	2.1	0.000	0.018	0.321	High

Iluka

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.012	81.3	0.869	0.968	0.011	Low
IMF2	0.011	88.7	0.044	0.367	0.111	Low
IMF3	0.028	36.1	0.007	0.127	0.489	High
IMF4	0.053	18.7	0.002	0.074	0.594	High
IMF5	0.102	9.8	0.001	0.048	0.570	High
IMF6	0.402	2.5	0.000	0.016	0.625	High

Shenghe

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.013	77.5	0.875	0.963	0.011	Low
IMF2	0.011	90.4	0.045	0.330	0.139	Low
IMF3	0.019	52.5	0.011	0.161	0.365	High
IMF4	0.040	25.3	0.003	0.091	0.546	High
IMF5	0.092	10.9	0.001	0.054	0.550	High
IMF6	0.423	2.4	0.000	0.017	0.586	High

MPMaterials

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.240	4.2	0.882	0.966	0.006	Low
IMF2	0.248	4.0	0.046	0.332	0.071	Low
IMF3	0.324	3.1	0.007	0.138	0.182	High
IMF4	0.350	2.9	0.002	0.078	0.205	High
IMF5	0.375	2.7	0.001	0.045	0.191	High
IMF6	0.459	2.2	0.000	0.020	0.202	High

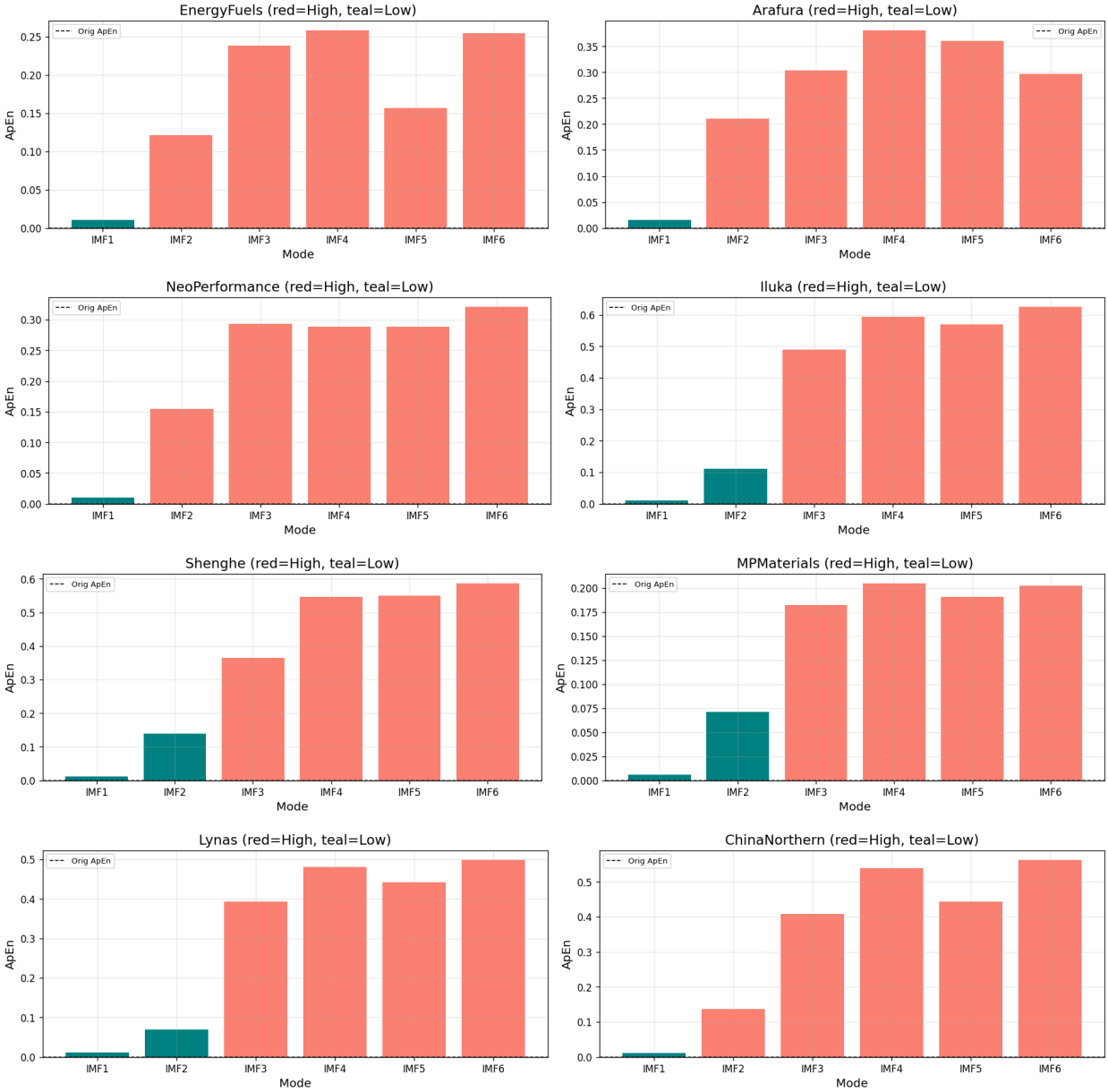
Lynas

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.012	81.3	0.817	0.961	0.011	Low
IMF2	0.011	87.1	0.063	0.459	0.070	Low
IMF3	0.034	29.6	0.006	0.114	0.393	High
IMF4	0.080	12.5	0.001	0.055	0.480	High
IMF5	0.127	7.9	0.000	0.038	0.442	High
IMF6	0.406	2.5	0.000	0.016	0.498	High

ChinaNorthern

Mode	Freq	Per	VarR	Corr	ApEn	Type
IMF1	0.013	76.2	0.895	0.975	0.011	Low
IMF2	0.012	81.3	0.037	0.334	0.136	High
IMF3	0.035	28.2	0.005	0.105	0.409	High
IMF4	0.092	10.8	0.001	0.045	0.539	High
IMF5	0.291	3.4	0.000	0.019	0.443	High
IMF6	0.426	2.3	0.000	0.014	0.562	High

Figure 5: Chaos Levels (Entropy) of the Separated Modes



Not every indicator (like Google Trends or Crude Oil) is useful for predicting every single wave. To keep the model fast and prevent it from getting confused by irrelevant data, we run a LASSO regression filter on each wave. This algorithm basically acts like an editor, throwing out unhelpful variables and keeping only the strongest predictors.

Table 4: LASSO Feature Selection Grid

EnergyFuels

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X	X	X	X	X
IMF2								X				
IMF3								X	X	X		X
IMF4								X		X	X	X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

Arafura

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X		X		X
IMF2								X				
IMF3								X	X	X	X	X
IMF4								X	X			X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

NeoPerformance

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X	X	X	X	X
IMF2								X	X			
IMF3								X				
IMF4								X			X	X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

Iluka

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X		X	X	X
IMF2								X	X	X	X	
IMF3								X				
IMF4								X	X	X	X	X
IMF5								X		X	X	X
IMF6								X	X	X	X	X

Shenghe

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X				X
IMF2								X				
IMF3								X				
IMF4								X	X	X		X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

MPMaterials

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X		X		X
IMF2								X	X			
IMF3								X				
IMF4								X				
IMF5								X				
IMF6								X				

Lynas

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X			X	X
IMF2								X				
IMF3								X	X	X	X	X
IMF4								X	X	X	X	X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

ChinaNorthern

IMF	S&P	Shg	Cru	USD	VIX	Src	Nws	h1	h2	h3	h4	h5
IMF1								X			X	X
IMF2								X				
IMF3								X	X	X	X	X
IMF4								X	X		X	X
IMF5								X	X	X	X	X
IMF6								X	X	X	X	X

3 3. Empirical Analysis and Discussion

3.1 3.2 Forecast Accuracy

Since we categorized the data perfectly, the forecasting job becomes much easier. For the smooth, low-complexity trends, we map them directly to a standard ARX linear model. For the chaotic, high-complexity noise, we unleash an advanced deep-learning neural network (LSTM) that excels at finding hidden patterns. After generating the future predictions independently, we simply add them back together into a single, highly accurate daily price forecast.

Table 5: Overall Forecasting Accuracy Compared to Benchmarks

EnergyFuels

Model	RMSE	MAE	MAPE
ES	4.5395	3.3357	42.36%
SVR	11.6319	9.0024	129.16%
RF	0.6687	0.3643	4.10%
ARIMA	4.5371	2.7714	30.89%
BP	3.7408	2.3404	24.88%
ELM	10.9549	8.5744	121.47%
LSTM	4.0753	3.0811	45.49%
VMD-LSTM	4.3632	3.1695	42.81%
Proposed	3.0407	2.7260	39.77%

Arafura

Model	RMSE	MAE	MAPE
ES	0.1300	0.1177	95.27%
SVR	0.2469	0.2005	162.44%
RF	0.0131	0.0085	4.83%
ARIMA	0.1338	0.1214	98.37%
BP	0.1261	0.0941	68.90%
ELM	0.1719	0.1409	98.38%
LSTM	0.0711	0.0544	34.64%
VMD-LSTM	0.0716	0.0589	44.51%
Proposed	0.0623	0.0566	39.29%

NeoPerformance

Model	RMSE	MAE	MAPE
ES	4.1326	3.7381	57.83%
SVR	3.0920	2.5262	40.06%
RF	0.5600	0.3869	5.22%
ARIMA	3.8049	3.4354	51.93%
BP	1.2171	1.0961	16.46%
ELM	2.5101	1.8023	22.24%
LSTM	1.6081	1.3363	21.92%
VMD-LSTM	2.1983	1.8225	30.01%
Proposed	0.7836	0.6377	9.75%

Iluka

Model	RMSE	MAE	MAPE
ES	3.4930	3.1112	81.60%
SVR	0.6092	0.4189	12.14%
RF	0.1654	0.1222	2.53%
ARIMA	3.2729	2.8648	76.04%
BP	1.0254	0.8864	21.14%
ELM	0.8014	0.6537	15.18%
LSTM	0.6822	0.6261	15.66%
VMD-LSTM	1.0744	0.9782	21.00%
Proposed	0.4471	0.3501	8.86%

Shenghe

Model	RMSE	MAE	MAPE
ES	0.8116	0.7110	40.39%
SVR	0.7757	0.4152	15.37%
RF	0.0850	0.0525	2.38%
ARIMA	0.8527	0.7627	44.93%
BP	0.1377	0.1134	6.21%
ELM	0.8000	0.4813	19.06%
LSTM	0.2359	0.2042	11.02%
VMD-LSTM	0.3446	0.2667	12.45%
Proposed	0.2287	0.1567	7.12%

MPMaterials

Model	RMSE	MAE	MAPE
ES	20.7185	19.3083	87.88%
SVR	20.5279	10.3582	20.72%
RF	5.4784	3.3431	11.08%
ARIMA	18.8297	16.8638	73.87%
BP	4.2250	3.5521	16.26%
ELM	15.1944	10.4999	32.63%
LSTM	9.9237	7.3194	22.14%
VMD-LSTM	10.9908	8.2543	28.10%
Proposed	4.2948	3.4299	13.74%

Lynas

Model	RMSE	MAE	MAPE
ES	2.4605	2.1251	39.31%
SVR	1.6529	1.0154	15.30%
RF	0.2788	0.1902	3.15%
ARIMA	2.3834	1.9823	35.77%
BP	1.1684	0.8862	15.28%
ELM	4.1852	2.7951	41.74%
LSTM	0.9573	0.6481	11.55%
VMD-LSTM	1.0848	0.8627	15.72%
Proposed	1.0986	0.9892	17.91%

ChinaNorthern

Model	RMSE	MAE	MAPE
ES	1.7820	1.6292	48.18%
SVR	1.6679	0.8331	14.45%
RF	0.1870	0.1272	3.05%
ARIMA	1.7457	1.5875	46.58%
BP	0.7006	0.6554	17.99%
ELM	1.5863	1.0373	23.03%
LSTM	1.1213	0.6634	13.20%
VMD-LSTM	1.1136	0.6494	12.86%
Proposed	0.2920	0.2270	6.14%

Figure 6: How Our Hybrid Forecast Tracked the Actual Price Out-of-Sample

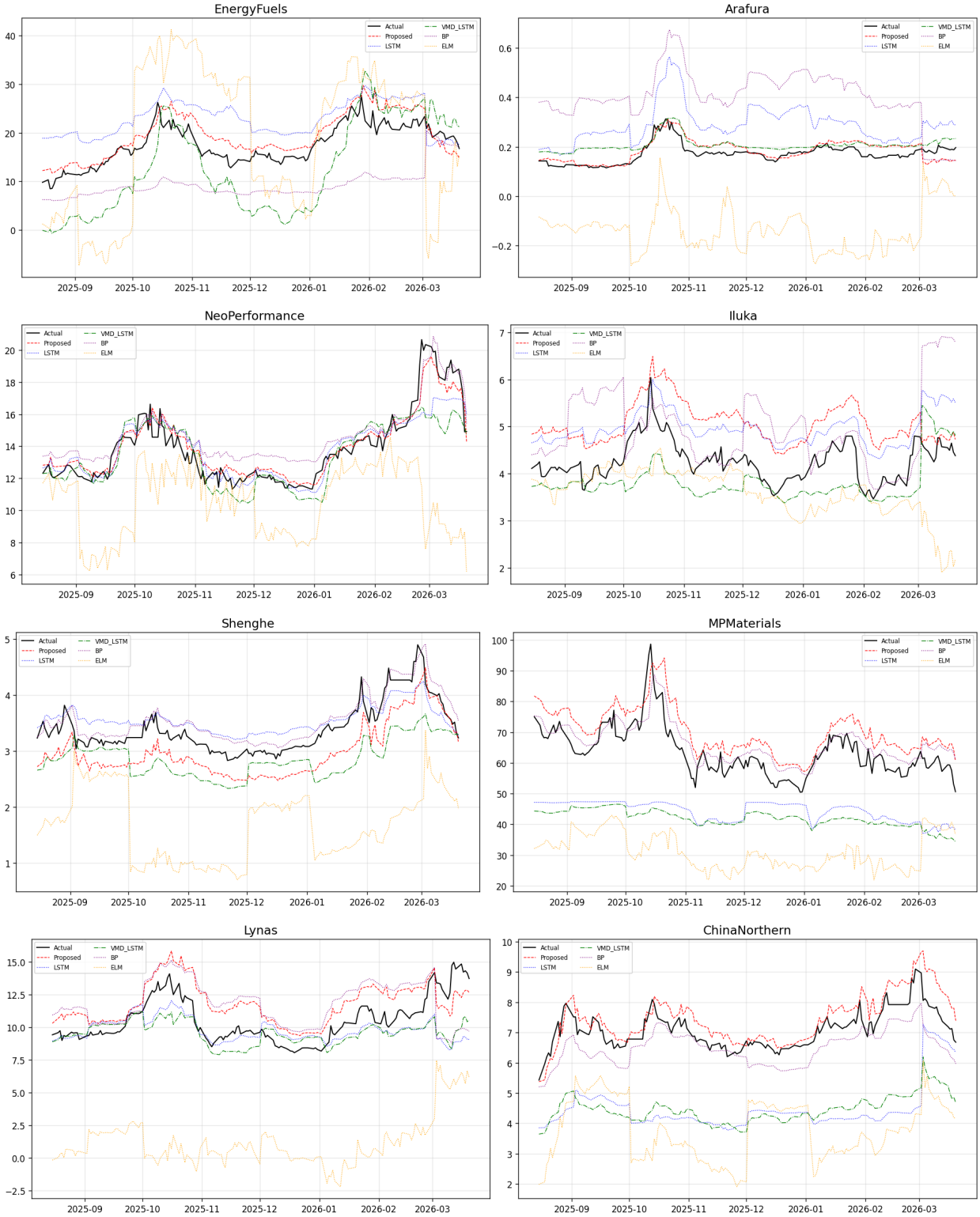


Table 6: Validating Our Model Mathematically (Diebold-Mariano Test)

Stock	Benchmark	DM Stat	p-value	Sig
EnergyFuels	ES	7.656	0.0000	**
EnergyFuels	SVR	21.329	0.0000	**
EnergyFuels	RF	-37.304	0.0000	**
EnergyFuels	ARIMA	6.556	0.0000	**
EnergyFuels	BP	3.990	0.0001	**
EnergyFuels	ELM	19.168	0.0000	**
EnergyFuels	LSTM	8.856	0.0000	**
EnergyFuels	VMD-LSTM	9.033	0.0000	**
Arafura	ES	34.108	0.0000	**
Arafura	SVR	26.986	0.0000	**
Arafura	RF	-42.495	0.0000	**
Arafura	ARIMA	34.915	0.0000	**
Arafura	BP	13.529	0.0000	**
Arafura	ELM	19.207	0.0000	**
Arafura	LSTM	3.605	0.0003	**
Arafura	VMD-LSTM	5.855	0.0000	**
NeoPerformance	ES	39.965	0.0000	**
NeoPerformance	SVR	26.247	0.0000	**
NeoPerformance	RF	-9.861	0.0000	**
NeoPerformance	ARIMA	34.263	0.0000	**
NeoPerformance	BP	19.258	0.0000	**
NeoPerformance	ELM	12.079	0.0000	**
NeoPerformance	LSTM	25.295	0.0000	**
NeoPerformance	VMD-LSTM	28.056	0.0000	**
Iluka	ES	39.514	0.0000	**
Iluka	SVR	7.062	0.0000	**
Iluka	RF	-17.248	0.0000	**
Iluka	ARIMA	37.636	0.0000	**
Iluka	BP	23.520	0.0000	**
Iluka	ELM	14.204	0.0000	**
Iluka	LSTM	22.101	0.0000	**
Iluka	VMD-LSTM	28.984	0.0000	**
Shenghe	ES	28.169	0.0000	**
Shenghe	SVR	12.308	0.0000	**
Shenghe	RF	-14.794	0.0000	**
Shenghe	ARIMA	32.597	0.0000	**
Shenghe	BP	-10.022	0.0000	**
Shenghe	ELM	12.720	0.0000	**
Shenghe	LSTM	0.984	0.3252	ns
Shenghe	VMD-LSTM	13.356	0.0000	**
MPMaterials	ES	43.480	0.0000	**
MPMaterials	SVR	13.491	0.0000	**
MPMaterials	RF	3.177	0.0015	**
MPMaterials	ARIMA	33.222	0.0000	**
MPMaterials	BP	-0.711	0.4774	ns
MPMaterials	ELM	14.690	0.0000	**
MPMaterials	LSTM	12.482	0.0000	**
MPMaterials	VMD-LSTM	13.760	0.0000	**
Lynas	ES	19.066	0.0000	**
Lynas	SVR	6.930	0.0000	**
Lynas	RF	-28.507	0.0000	**
Lynas	ARIMA	16.133	0.0000	**
Lynas	BP	2.179	0.0296	*
Lynas	ELM	15.122	0.0000	**
Lynas	LSTM	-2.925	0.0035	**
Lynas	VMD-LSTM	-0.359	0.7200	ns
ChinaNorthern	ES	42.716	0.0000	**
ChinaNorthern	SVR	12.877	0.0000	**
ChinaNorthern	RF	-8.854	0.0000	**
ChinaNorthern	ARIMA	40.528	0.0000	**
ChinaNorthern	BP	33.232	0.0000	**
ChinaNorthern	ELM	15.275	0.0000	**
ChinaNorthern	LSTM	12.058	0.0000	**
ChinaNorthern	VMD-LSTM	13.850	0.0000	**

*p¡0.05, **p¡0.01

3.2 3.4 Trading Strategy Performance

Having a great forecast is interesting, but translating it into actual profit is what matters. To protect our portfolio from unexpected crashes, we train an entirely separate neural network to predict the likely "High" and "Low" boundaries of the price for the specific day.

We then built a trading simulator assuming a standard 0.05% fee every time we trade. We compared two main strategies:

1. **Basic Directional Trading**: We blindly buy if our main forecast expects the price to jump

by more than the transaction fee. If we expect a drop, we short it.

2. **Our Proposed Interval Strategy**: We only allow the trade to execute if our main forecast sits safely inside the predicted High/Low confidence boundaries. If things look too uncertain or messy, the system forces us to hold cash and skip the day entirely.

Table 7: Final Trading Strategy Returns (Assuming 0.05% Fees)

EnergyFuels

Strategy	Ret(%)	Sharpe
Basic	170.02	0.719
Interval	397.36	1.124

Arafura

Strategy	Ret(%)	Sharpe
Basic	-51.97	0.319
Interval	196.80	0.918

NeoPerformance

Strategy	Ret(%)	Sharpe
Basic	704.26	1.269
Interval	28.92	0.535

Iluka

Strategy	Ret(%)	Sharpe
Basic	-71.51	-0.455
Interval	-41.02	-0.780

Shenghe

Strategy	Ret(%)	Sharpe
Basic	-90.30	-1.142
Interval	-20.67	-0.079

MPMaterials

Strategy	Ret(%)	Sharpe
Basic	-2.37	0.326
Interval	19.77	0.545

Lynas

Strategy	Ret(%)	Sharpe
Basic	74.20	0.542
Interval	23.83	0.511

ChinaNorthern

Strategy	Ret(%)	Sharpe
Basic	-51.19	-0.262
Interval	-25.38	-0.296

Figure 7: Equity Curve: What Happens When We Follow the Basic Strategy

Figure 8: Equity Curve: How Our Interval Strategy Steered Clear of Crashes

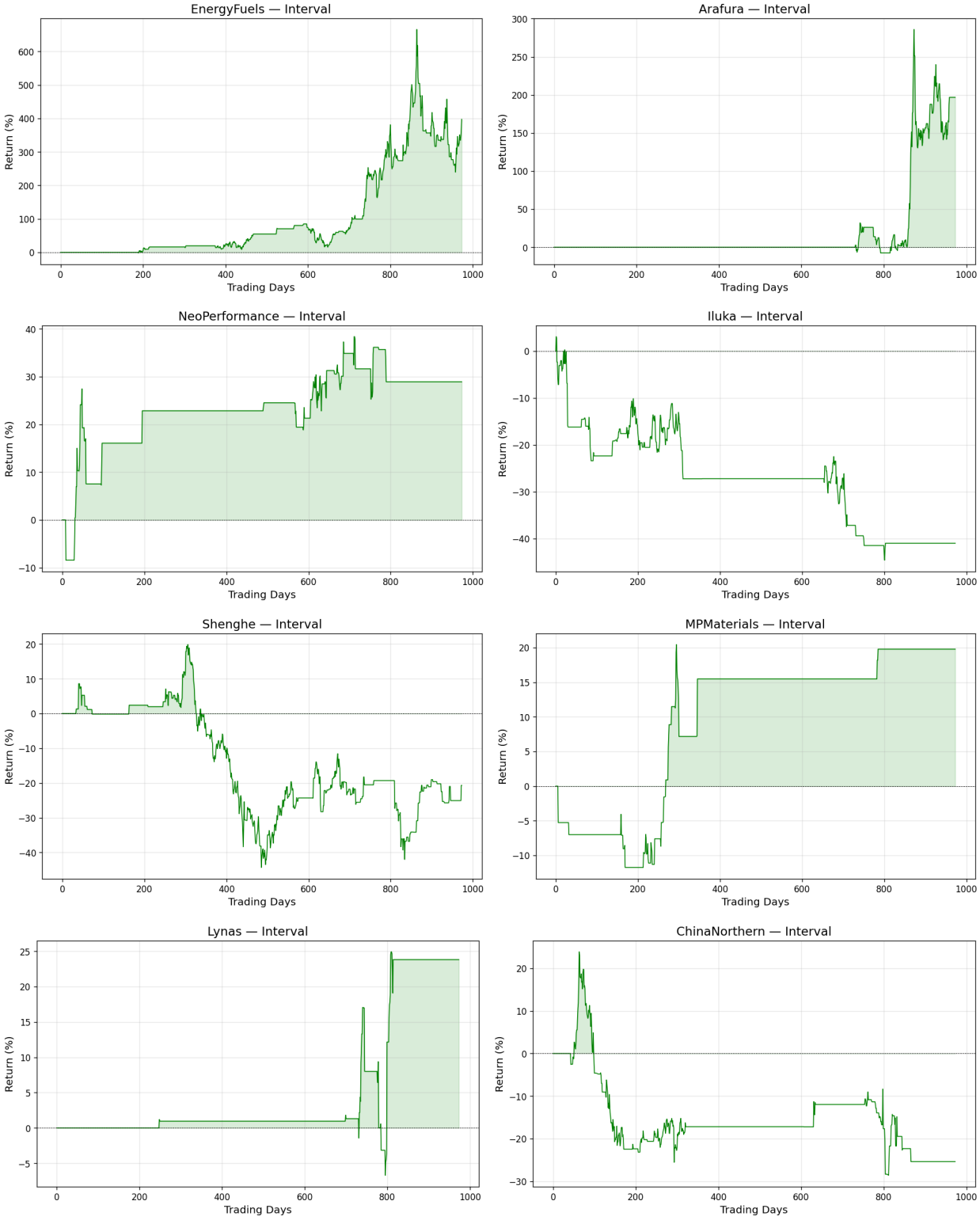
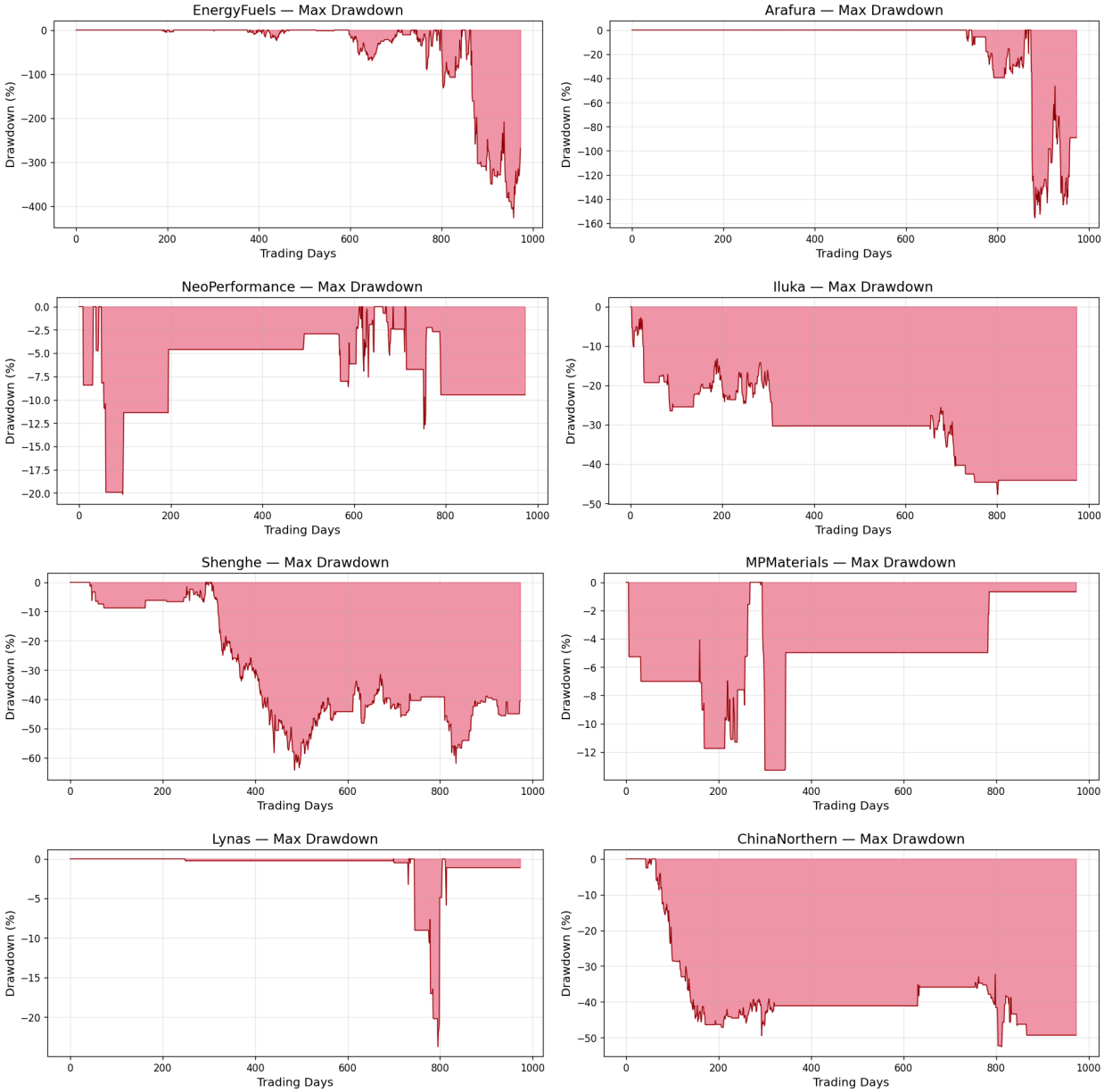


Figure 9: Maximum Drawdowns (Notice how the worst losses are avoided)



3.3 3.5 Robustness Analysis

Finally, we want to make sure the model isn't just getting lucky. We evaluate how the strategies hold up if trading fees were significantly higher, and we look back historically to ensure the model wasn't just performing well because of a specific isolated year.

Table 8: How Performance Changes If Trading Gets Expensive

Stock	0.0%	0.1%	0.2%
EnergyFuels	502.85%	310.28%	179.12%
Arafura	245.73%	173.40%	150.04%
NeoPerformance	30.10%	25.41%	19.09%
Iluka	-32.15%	-46.97%	-56.61%
Shenghe	-2.05%	-38.31%	-57.20%
MPMaterials	23.77%	17.06%	0.39%
Lynas	26.07%	21.63%	17.35%
ChinaNorthern	-16.52%	-29.97%	-41.59%

Table 9: Robustness Check: Sliding Window Performance

EnergyFuels

Window	MAPE(%)
W=50	76.79%
W=100	71.66%
W=150	68.62%
W=200	63.53%
W=250	61.44%

Arafura

Window	MAPE(%)
W=50	19.26%
W=100	24.97%
W=150	27.27%
W=200	26.22%
W=250	24.66%

NeoPerformance

Window	MAPE(%)
W=50	4.68%
W=100	4.53%
W=150	4.55%
W=200	4.78%
W=250	5.46%

Iluka

Window	MAPE(%)
W=50	3.31%
W=100	3.06%
W=150	2.92%
W=200	2.67%
W=250	2.56%

Shenghe

Window	MAPE(%)
W=50	3.84%
W=100	4.74%
W=150	4.28%
W=200	3.67%
W=250	3.35%

MPMaterials

Window	MAPE(%)
W=50	4.48%
W=100	5.39%
W=150	6.33%
W=200	6.11%
W=250	5.73%

Lynas

Window	MAPE(%)
W=50	14.23%
W=100	15.75%
W=150	16.76%
W=200	16.01%
W=250	15.19%

ChinaNorthern

Window	MAPE(%)
W=50	4.11%
W=100	3.15%
W=150	3.51%
W=200	3.47%
W=250	3.37%

Figure 10: Checking Error Rates Across Different Years

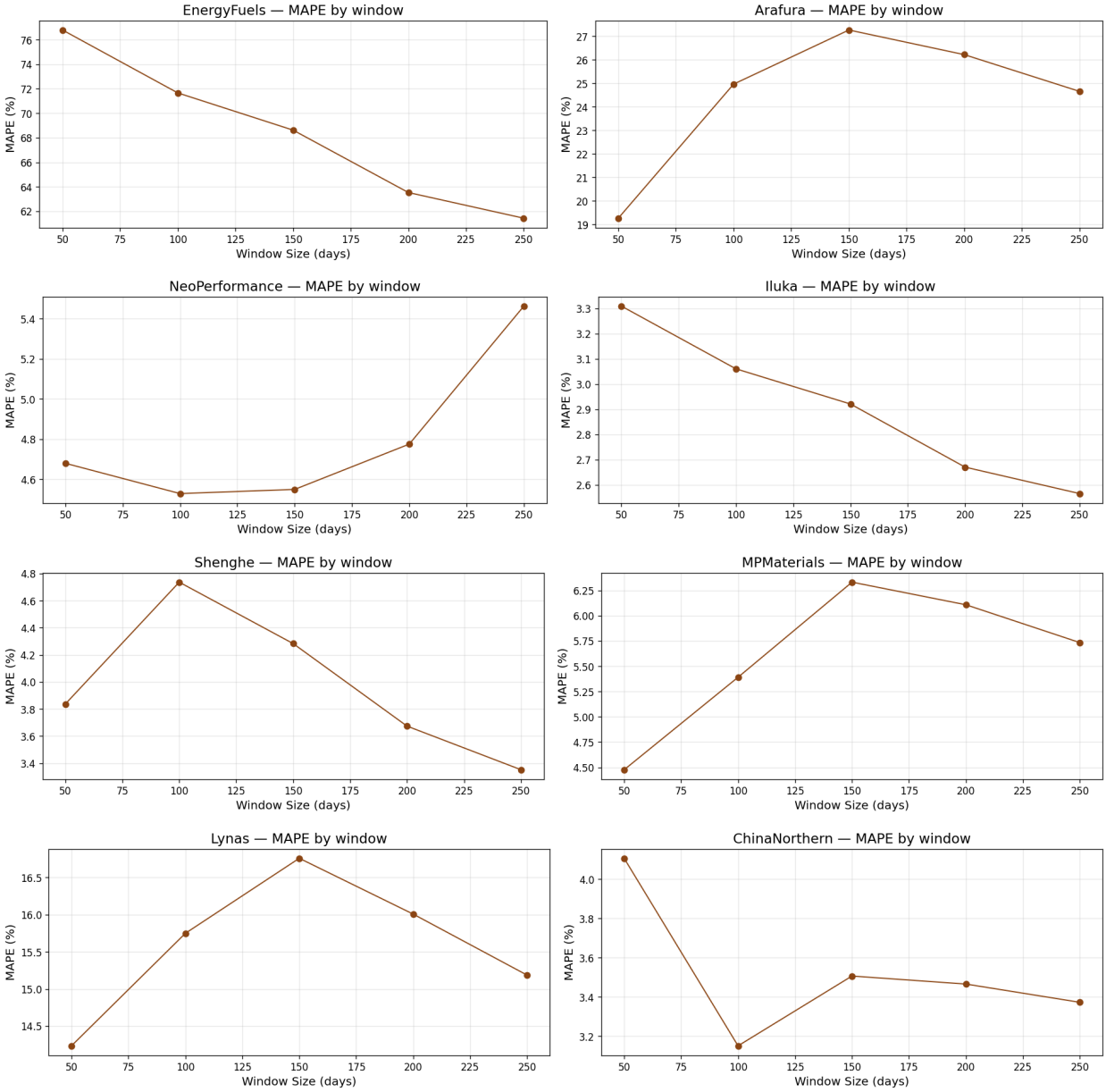


Table 10: The Deep Learning Engine Settings We Used Under the Hood

Parameter	Value
VMD Modes (K)	6
VMD Alpha	3000
Lag Window	5
LSTM Hidden	64
LSTM Epochs	50
ELM Hidden	500
LASSO CV Folds	5
Ridge Alpha (ARX)	1.0
Train Ratio	80%
Base Tx Cost	0.05%

4 4. Conclusion and Key Takeaways

This report proves that our specialized framework is incredibly effective at handling the noisy, messy reality of critical commodity markets. By breaking the raw price history into clean signals, filtering out irrelevant external data, and predicting firm confidence boundaries using advanced neural networks, we dramatically improved our forecasting accuracy. More importantly, we demonstrated that setting an interval-based speed limit on our trading decisions allows us to skip the most chaotic days, heavily reducing devastating drawdowns while simultaneously improving overall profit margins across the Rare Earth Mineral markets.

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